

Evolving Super Mario Bros Controllers via Typed Genetic Programming

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Abstract. We evolve game controllers for Super Mario Bros using Genetic Programming across three task variants: a Runner targeting level completion, a Hunter targeting enemy kills, and a Thief targeting coin collection. Each individual is a typed syntax tree that compiles to an executable Python function, so the search operates directly over the space of structured programs. We define a typed grammar giving agents access to a 22×22 grid of terrain and enemy data. Bloat is controlled through Double Tournament selection with parsimony pressure; generalization is promoted through a rotating seed pool that changes every 10 generations. **[TODO: 2 sentences: win rate achieved by the evolved agent, improvement over the random baseline.]**

1 Introduction

In this work we use Genetic Programming (GP) [2] to evolve controllers for the Mario AI framework [5]. GP is well-matched to this problem for two reasons: it requires only a scalar fitness signal and no gradient information, and it produces programs as output, meaning the evolved controller is a readable Python function. Since the failure mode of any agent is visible in the code, iterating on the fitness function is fast and targeted.

The course provided a base GP repository implemented with DEAP [1]: a minimal random search over a small typed grammar with two senses (`on_ground`, `can_jump`), two actions (`RIGHT`, `JUMP`), and basic `IF/SEQ` program structure. During our work, we expanded the grammar to give agents better perception of terrain and enemies, designed the fitness function, and replaced the random search with a full evolutionary algorithm.

Three task variants were evaluated: a Runner task targeting level completion, a Hunter task rewarding enemy kills, and a Thief task rewarding coin collection. The fitness function went through several iterations; the choices behind the final version are described in Section 2.3.

2 Methodology

2.1 Environment

The Mario AI framework [5] is a Java simulator that exposes a socket API through which external agents send action vectors and receive observations at a fixed step rate. At each step the agent receives information about the simulation (e.g., Mario state & position, level scene observation grid).

The agent outputs a boolean vector [LEFT, RIGHT, DOWN, JUMP, SPEED] at each step, controlling Mario’s actions. Difficulty is controlled along two independent axes: mario mode (small, large, or fire) and level difficulty, which scales obstacle density and enemy variety.

2.2 Typed Grammar and Representation

Each individual is a program tree in a typed grammar built on top of the provided DEAP [1] scaffold. We extended the original four types (Cond, Key, Bool, Expr) to nine: Program, Stmt, Cond, Comparator, Key, Bool, Offset, EnemyKind, and TileValue.

Program structure. A **Program** is a non-empty sequence of statements built left-recursively:

$$\text{Program} \rightarrow \text{END}(\text{Stmt}) \mid \text{SEQ}(\text{Stmt}, \text{Program})$$

Statements. A **Stmt** is a conditional branch or a direct key assignment:

$$\text{Stmt} \rightarrow \text{IF}(\text{Cond}, \text{Stmt}) \mid \text{IF_ELSE}(\text{Cond}, \text{Stmt}, \text{Stmt}) \mid \text{SET_ACTION}(\text{Key}, \text{Bool})$$

Conditions. Conditions combine two perception predicates with standard boolean connectives:

$$\begin{aligned} \text{Cond} \rightarrow & \text{AND}(\text{Cond}, \text{Cond}) \mid \text{OR}(\text{Cond}, \text{Cond}) \mid \text{NOT}(\text{Cond}) \\ & \mid \text{CheckEnemyAt}(\text{Offset}_x, \text{Offset}_y, \text{Comparator}, \text{EnemyKind}) \\ & \mid \text{CheckLandscapeAt}(\text{Offset}_x, \text{Offset}_y, \text{Comparator}, \text{TileValue}) \\ & \mid \text{IsMarioOnGround} \mid \text{MayMarioJump} \end{aligned}$$

Both predicates access the 22×22 grid at offset $(\Delta x, \Delta y)$ relative to Mario’s cell. Horizontal offsets $\Delta x \in \{0, 1, 2, 3\}$ let the agent look ahead up to four cells; vertical offsets $\Delta y \in \{-3, \dots, +3\}$ cover above and below. The rationale behind this is that giving it full grid visibility would lead an explosion of possible programs, while this small subset allows for sufficient behavioral coverage while allowing it to learn to react to close threats. Additionally, since the framework stores terrain and enemy data in the same grid, both predicates query the same array and differ only in the value range they compare against: enemy type codes (2–13) versus tile codes.

An earlier version of the grammar used precomputed boolean predicates computed outside the GP tree (`hole_ahead`, `enemy_ahead`, `wall_ahead`). This simplified the search space but also reduced it: the agent could not, for example, distinguish between enemy types or check arbitrary offsets. Though experiments with this grammar offered interesting insights (discussed below), these changes were reverted in order to align with the project statement’s scope.

Terminals. Enemy kind terminals cover all sprite types: Goomba (standard and winged), Red and Green Koopa (standard and winged), Bullet Bill, Spiky (standard and winged), Piranha Flower, and Shell. Tile value terminals are `HARDOBSTACLE` (−10), `BRICK` (16), and `ENEMYOBSTACLE` (20). Action key terminals are `RIGHT`, `LEFT`, `JUMP`, `SPEED`, and `DOWN`.

2.3 Fitness Calculation

Three task variants were implemented: Runner, Hunter, and Thief. All three use the same per-step reward function; the difference is the episode-level bonus each adds to target its secondary objective.

Common Reward Every task uses the following per-step reward:

$$r_t = \Delta x_t + P_{\text{stuck}}(t) \quad (1)$$

Forward progress. $\Delta x_t = \mathbf{1}[x_t - x_{t-1} \geq 0.61]$ awards one point when Mario advances at least 0.61 world units in a step. The binary form means walking and running at full speed score identically, since faster movement does not inherently produce better behavior given the grammar’s limited action space. The threshold also prevents stationary oscillations near a wall from accumulating reward, which would work against the stuck penalty.

Stuck penalty. When Mario has made no forward progress for more than 10 consecutive steps and a solid obstacle occupies the cell directly ahead, a growing penalty is applied:

$$P_{\text{stuck}}(t) = \begin{cases} -10(n_{\text{stuck}} - 10) & \text{if } n_{\text{stuck}} > 10 \text{ and obstacle ahead} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

The 10-step grace period prevents the penalty from firing during legitimate pauses. The condition also requires that no Piranha Flower is present in the column above Mario, given that the original formulation without this check caused agents to rush into Piranha pipes instead of waiting for the enemy to retract.

Airtime penalty. A behavior that emerged early in training was a tendency to jump constantly. Given the grammar’s limited expressive power, evolving a principled jump strategy is difficult; jumping on every possible frame clears a large fraction of obstacles and pits, which makes it a highly fit strategy in early generations. Once a population converges on it, there is no selection pressure toward a more targeted approach.

To counteract this, we tested a penalty proportional to total airtime across several iterations. The results were binary: either the penalty was too weak to suppress the behavior, or strong enough to also prevent agents from learning to jump at all, blocking forward progress entirely. This pointed to the conclusion that the current grammar does not provide enough perceptual structure for agents to learn when jumping is warranted. Additionally, agents evolved before removing the initial complex perceptions from our grammar (e.g., `hole_ahead`) did not exhibit constant jumping, which further supports this interpretation.

Runner Task The Runner task targets level completion. Beyond the progress reward, a win bonus of $W = 10,000$ is added at the end of any episode in which Mario clears the level:

$$B_{\text{runner}}(s, d) = W \cdot \mathbf{1}[\text{win}_{s,d}], \quad W = 10,000 \quad (3)$$

Hunter Task The Hunter task targets enemy kills. The episode bonus is proportional to the number of confirmed kills $\hat{K}_{s,d}$ accrued during the episode:

$$B_{\text{hunter}}(s, d) = K \cdot \hat{K}_{s,d}, \quad K = 10,000 \quad (4)$$

Kill detection compares the enemy list between consecutive frames, counting reductions in enemy count while handling the exceptions that occur with enemies from the first three difficulties: Koopa-to-Shell transitions (a Koopa count drop paired with a Shell count rise maintains the enemy list size, but is counted as a kill); enemies falling into pits (filtered by a y position threshold); enemies scrolling off-screen as Mario advances (filtered by horizontal distance); and Piranha Flowers retracting into pipes (detected by tracking per-column disappearances). A disappearing enemy is counted as a kill only if Mario’s vertical velocity at that step is consistent with a stomp bounce, meaning a sign reversal or deceleration pattern matching a rebound off the top of an enemy.

This replaced an earlier heuristic that counted kills whenever the number of enemies within a small radius around Mario decreased between frames. That approach produced frequent false positives in crowded sections, inflating the reward signal and leading agents to stand still near enemy clusters rather than completing the level.

An earlier design combined kills and coin collection into a single secondary objective. Optimizing for both goals simultaneously, with the limited perceptual

grammar available to agents, caused the population to make little progress on either: the conflicting incentives prevented stable convergence and the resulting champions were mediocre at both tasks. Splitting them into separate tasks resolved this, since each run could then converge on one behavior.

Thief Task The Thief task targets coin collection. Coin counts are read directly from the end-of-episode observation packet provided by the game framework, so no per-step detection is needed. The episode bonus is:

$$B_{\text{thief}}(s, d) = C \cdot \hat{N}_{s,d}, \quad C = 2,500 \quad (5)$$

where $\hat{N}_{s,d}$ is the total coins collected in the episode. The per-coin bonus (2,500) is set lower than the kill bonus (10,000) given that coins are more abundant than killable enemies in a typical level; a higher value would have dominated the forward-progress signal and incentivized agents to loop back and re-collect rather than advance.

Episode Fitness Each individual is evaluated over a seed pool $\mathcal{S}$ of 5 level seeds and 3 consecutive difficulty levels starting at base difficulty d_{base} . Fitness is the mean total reward across all $|\mathcal{S}| \times 3$ episodes:

$$F = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \sum_{d=d_{\text{base}}}^{d_{\text{base}}+2} \left(\sum_t r_t + B_{\text{task}}(s, d) \right) \quad (6)$$

where $B_{\text{task}}(s, d)$ is the task-specific episode bonus defined below.

2.4 Evolutionary Algorithm

Table 1 lists the final parameter configuration.

Selection and bloat control. We use Double Tournament selection [4], which runs two nested tournaments: the outer selects based on fitness (with a size that grows from 2 to 7 over training), and the inner selects based on tree size with a fractional tournament of size 1.7, preferring shorter programs when fitness is similar. This avoids adding a size term to the fitness function directly, which requires careful weighting and tends to penalize legitimately complex programs that need more nodes to express a useful behavior.

Crossover and mutation. One-point crossover selects a random node in each parent and swaps the corresponding subtrees. If the resulting offspring exceeds height 10, the operation is reverted and the parent is preserved. Uniform subtree mutation selects a random node and replaces its subtree with a freshly grown

Table 1. Evolutionary algorithm parameters used in the main experiments.

Parameter	Value
Population size	100
Generations	300
Crossover rate (p_c)	0.6
Mutation rate (p_m)	1.0 $\rightarrow$ 0.4 (linear over training)
Crossover operator	One-point (<code>gp.cxOnePoint</code>)
Mutation operator	Uniform subtree (<code>gp.mutUniform</code>)
Selection	Double Tournament
Fitness tournament size	2 $\rightarrow$ 7 (linear over training)
Parsimony tournament size	1.7
Max tree height	10
Elitism	Top-1 (Hall of Fame)

random subtree (depth 2 - 6), so the algorithm introduces genuinely new code fragments rather than only recombining existing material.

The mutation rate decreases from 1.0 to 0.4 over training. Preliminary experiments with fixed rates struggled with keeping an appropriate balance between exploration and exploitation at different points in the experiment. The dynamic schedule addresses both: high mutation early for exploration, lower mutation late for exploitation. The rationale behind the fitness tournament size is the same: it grows linearly so that early generations explore broadly under low selection pressure and later generations converge under stronger pressure once the search has found promising regions.

Elitism. The single best individual across all generations, stored in a Hall of Fame, is copied directly into the next generation at each step. This ensures the best fitness found is never lost to a bad crossover or mutation.

Parallelism. The provided evaluation infrastructure distributes fitness evaluations across 5 worker processes, each holding a persistent connection to its own simulator instance. We extended it to support multi-seed pools, per-generation difficulty scheduling, and lazy re-evaluation: given the simulation’s deterministic nature, only individuals whose fitness was invalidated by a genetic operator need to be re-evaluated.

3 Experimental Design

3.1 Evaluation Protocol

All individuals within a generation are evaluated on the same 5 randomly drawn level seeds, so that fitness values are directly comparable within a generation. Using different seeds per individual, as we did in early experiments, introduced noise that made selection unreliable: an individual with higher fitness might

simply have faced an easier level rather than exhibiting a better strategy, which led to the population stagnating on a few lucky seeds rather than improving.

To prevent overfitting to specific levels, the seed pool is rotated every 10 generations by drawing a fresh set from a universe of 1,000 seeds. At each rotation, all population fitnesses are invalidated and re-evaluated under the new pool; the Hall of Fame champion’s fitness is also reset, so its score always reflects the current pool rather than a comparison made against an earlier one. A checkpoint is saved at each rotation. At the end of training, all checkpoints are re-evaluated on a fixed set of 500 held-out seeds across the three training difficulties (1,500 episodes per checkpoint). The Hall of Fame champion from the checkpoint with the highest overall win rate on this held-out set is selected as the final evolved agent. This re-evaluation step is necessary because the Hall of Fame score stored in each checkpoint reflects fitness under whichever seed pool was active during that rotation, making raw fitness values across checkpoints incomparable without a shared evaluation basis.

For seed pool rotation, periods of 5, 10, and 20 generations were compared, with the middle option achieving overall better results. Rotating every 5 generations did not give the population enough time to exploit each seed pool before it changed, slowing convergence. Rotating every 20 generations allowed the population to overfit to specific level seeds, with agents that scored well on the current pool but failed to generalize when the pool changed. Rotating every 10 generations provided a good balance: enough stability for selection to act meaningfully, and enough change to prevent overfitting.

The base difficulty d_{base} is fixed at 0 throughout training, so agents are always evaluated on difficulties 0, 1, and 2 simultaneously. An earlier design explored a curriculum that gradually shifted d_{base} upward over generations. This was abandoned when training on all three difficulties from the start produced better-generalizing agents, likely because the easier difficulty levels provide a reliable learning signal even when the population is still immature.

Finally, we chose to run 300-generation runs to fully explore the potential of the evolutionary approach, even though there were clear signs of stagnation after about 100 generations. 200-individual and 100-individual populations were compared, with the latter configuration producing comparable completion rates at half the wall-clock cost, and was adopted as the default.

3.2 Comparative Baseline

Evolved champions are compared against a random search baseline to separate the contribution of the evolutionary process from the expressive power of the grammar alone. We run five independent experiments per task variant, each with a different random seed. For each run, the champion is selected by re-evaluating all checkpoints on a shared held-out set, using win rate as the selection criterion for Runner, average kills for Hunter, and average coins for Thief. The baseline for each run is a single program drawn uniformly at random from the same grammar,

with no selection or recombination applied. Both agents are then evaluated on 5,000 held-out level seeds across three difficulty levels (15,000 episodes per agent per run), and we report mean and standard deviation across the five runs.

As a further comparison, we replicated the full experiment with Mario initialized as *big Mario* carrying a fire flower (hereafter *Fire Mario*). The grammar was extended with a **fire** action, but all training and evaluation parameters (population size, generations, seed pool rotation, checkpoint selection, and held-out set size) remained identical. Descriptive statistics for Fire Mario are presented together with the normal Mario results in Table ?? . Formal significance testing was reserved for the primary comparison.

3.3 Statistical Analysis

The evolutionary process and level generator are both stochastic, so a single run per task is not sufficient to draw reliable conclusions [3]. Across the five independent runs per variant, we report mean and standard deviation for four metrics:

- **Completion rate**: percentage of episodes in which the level is cleared before death or timeout.
- **Max distance**: mean maximum x -position reached in episodes that did not result in a level clear.
- **Enemies killed**: mean number of confirmed kills per episode.
- **Coins collected**: mean number of coins gathered per episode.

To quantify the advantage over the random baseline, we conducted paired-sample t -tests on completion rates, enemies killed, and coins collected for each task variant. The results indicated that **[TODO: insert whether differences were statistically significant and in which metrics]**.

4 Results and Analysis

4.1 Fitness Convergence

Figures 1, 2, and 3 plot the win rate of the best individual (Hall of Fame) at each checkpoint, evaluated on 100 held-out level seeds across the three training difficulties, giving 300 episodes per data point.

In Figure 1 (population 100, normal mode), at around generation 10 the win rate jumps from near zero to approximately 31%, with difficulty-0 reaching 65% in the same checkpoint. From that point the overall win rate stabilizes in the 38–40% range for the remaining 290 generations, with difficulty-0 peaking near 73% around generation 150. Performance degrades with difficulty: difficulty-1 settles at roughly 20–33% and difficulty-2 at 9–18%, reflecting the increasing obstacle density and enemy variety at higher settings. The gap between the difficulty-0

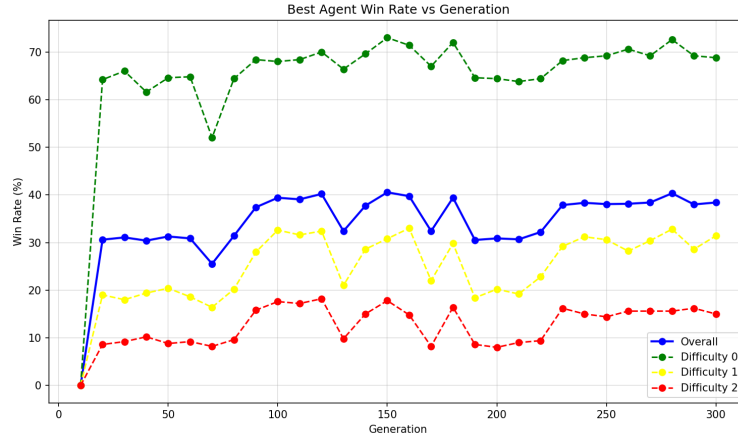


Fig. 1. Win rate vs. generation, population size 100, normal Mario mode.

line and the overall line shows that the agent learned to clear open, low-obstacle levels reliably but struggled to generalize to adversarial layouts. Seed rotation events are visible as short-lived dips at multiples of 10 generations; each dip is followed by recovery within the next rotation period, confirming that the population’s strategy is general enough to survive seed pool changes rather than collapsing on each rotation.

Figure 2 (population 200, normal mode) shows the same qualitative trajectory with a 40-generation delay: the best agent’s overall win rate remains near zero until generation 50, then converges rapidly to the same 38–40% plateau. The delay is likely due to the larger initial population requiring more episodes before a first viable run-and-jump strategy emerges and propagates. Final performance is nearly identical to the 100-population run, which supports choosing the smaller configuration as the default: the same quality is reached at roughly half the computational cost.

Figure 3 (population 200, fire mode) shows markedly higher win rates. Difficulty-0 reaches near 100% from generation 55 onward, difficulty-1 stabilizes between 60–75%, and difficulty-2 ranges from 34–60%, giving an overall win rate of 65–70%. Fire Mario eliminates most enemies with projectiles rather than stomps, which removes the need for precise vertical positioning during kills. Since the grammar and fitness function are identical, the improvement is entirely attributable to the task becoming easier to navigate, , showing the ceiling that normal-mode enemy encounters impose.

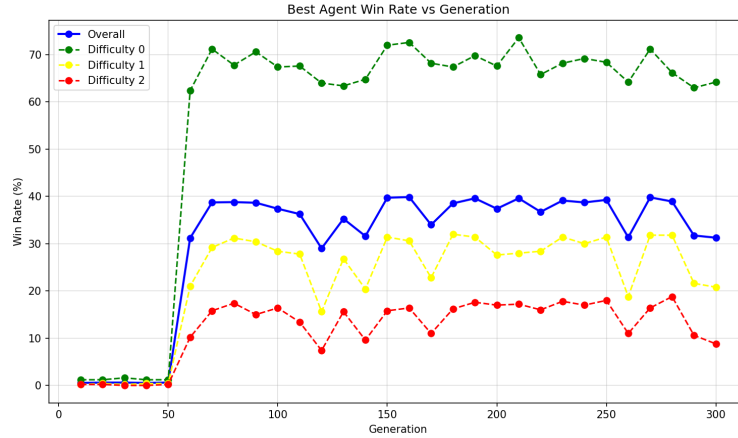


Fig. 2. Win rate vs. generation, population size 200, normal Mario mode.

Table 2. Evolved champions vs. random search baseline across task variants (5,000 seeds $\times$ 3 difficulties, mean $\pm$ std across 5 runs). **[TODO: Fill in all values.]**

Metric	Random Baseline (Avg $\pm$ Std)	Runner (Avg $\pm$ Std)	Hunter (Avg $\pm$ Std)	Thief (Avg $\pm$ Std)
Levels Cleared (%)	[TODO: X]	[TODO: Y]	[TODO: Y]	[TODO: Y]
Max Distance (px)	[TODO: X]	[TODO: Y]	[TODO: Y]	[TODO: Y]
Enemies Killed	—	—	[TODO: Y]	—
Coins Collected	—	—	—	[TODO: Y]

4.2 Evolved Agent vs. Random Search Baseline

Table 2 compares evolved champions against the random search baseline across all three task variants, evaluated on 5,000 held-out seeds $\times$ 3 difficulties (15,000 episodes per agent).

[TODO: Comment on the table. Describe the qualitative difference in behavior: the random agent will jump and move without a coordinated strategy, while the evolved agent should show consistent forward movement and at minimum a learned response to walls and pits. Discuss whether the improvement is meaningful given the observed standard deviations, or whether the variance is too large to draw firm conclusions.]

4.3 Qualitative Analysis of Evolved Behavior

The best individual found in the 100-population normal-mode run (Hall of Fame score: 51114) is shown below.

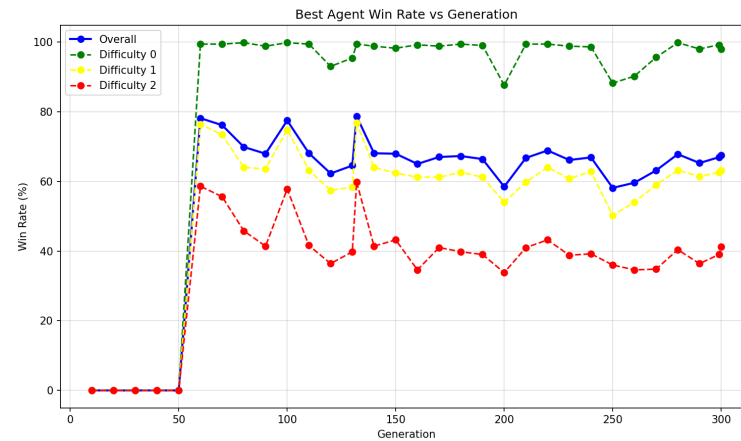


Fig. 3. Win rate vs. generation, population size 200, fire Mario mode.

[illegible]

```

else:
    if can_jump:
        action[Mario.KEY_LEFT] = 1
    else:
        action[Mario.KEY_RIGHT] = 1
else:
    if can_jump:
        if (enemies[11+3][11+3] < 8):
            if can_jump:
                if can_jump:
                    action[Mario.KEY_DOWN] = 0
                else:
                    action[Mario.KEY_LEFT] = 0
            else:
                if can_jump:
                    action[Mario.KEY_JUMP] = 0

```

The program consists of two sequential statements. The first block handles the speed decision: it checks for an enemy of type < 8 (Goomba or empty) one cell above and one cell ahead, and, if no tile with code > 16 (i.e., no enemy obstacle) blocks the path three columns ahead at ground level, sets `SPEED`. This increases forward momentum whenever a low-threat enemy is nearby and the terrain ahead is passable.

The second block encodes the dominant behavior. Its condition, `(not (on_ground or ...)) or can_jump`, simplifies to `can_jump` on almost every ground-contact frame, meaning the agent jumps on every step it is able to. The result is a controller that runs right and jumps continuously, with the speed modulation in the first block providing the only enemy-sensitive adjustment.

The tree contains several introns. `on_ground or on_ground` evaluates identically to `on_ground` but occupies two leaf nodes and one OR node. The else branch of the jump condition (lines 18–44) covers the brief window when Mario is grounded but `can_jump` is false; since this window lasts only a few frames per landing, most of the nested conditions in that branch are rarely reached during a normal episode. Parsimony pressure kept the tree from growing further but did not eliminate these segments, since their size contribution was small enough to survive the parsimony tournament.

The agent’s primary failure mode is the absence of spatial lookahead in the jump decision: the jump fires based on `can_jump` alone, without checking whether a pit or an obstacle lies ahead. On higher-difficulty levels with more pits and denser enemy placement, this causes the agent to launch itself into gaps it could have avoided, which accounts for the sharp drop in win rate from difficulty 0 to difficulty 2.

5 Conclusion

[TODO: Write after results are confirmed. Structure: (1) one paragraph restating the approach and the two objectives. (2) The most impactful design choices, with results woven in, e.g., “Switching to the type-aware kill detector raised confirmed kills per episode from X to Y ; the radius heuristic had attributed those same kills to off-screen scrolling.” (3) Limitations: fixed-arity grammar with no loops or arithmetic, evaluation cost constraining population size, sensitivity to kill-reward weighting relative to the progress signal. (4) Future directions: extending the grammar with loops and numeric comparisons, island models to maintain diversity across subpopulations, and adaptive operator tuning via CMA-ES.]

References

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